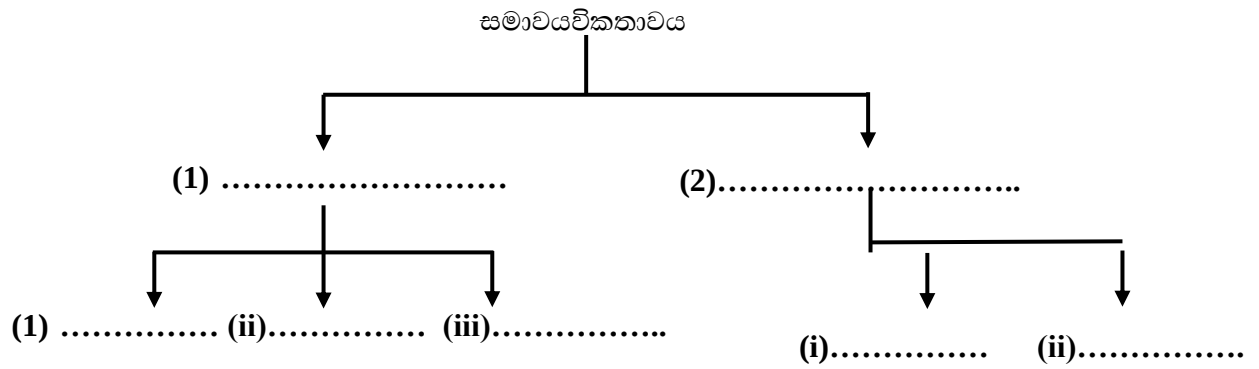


සමාවයවිකතාවය



(1):-

.....

(i)

හාශ්වා



හේට්ට්ඨාරච්චි

(ii)

ආරක්ෂා
සෞඛ්‍ය ආරක්ෂා



භාෂ්‍යා දේශීය ආරක්ෂාව



(2):-

.....

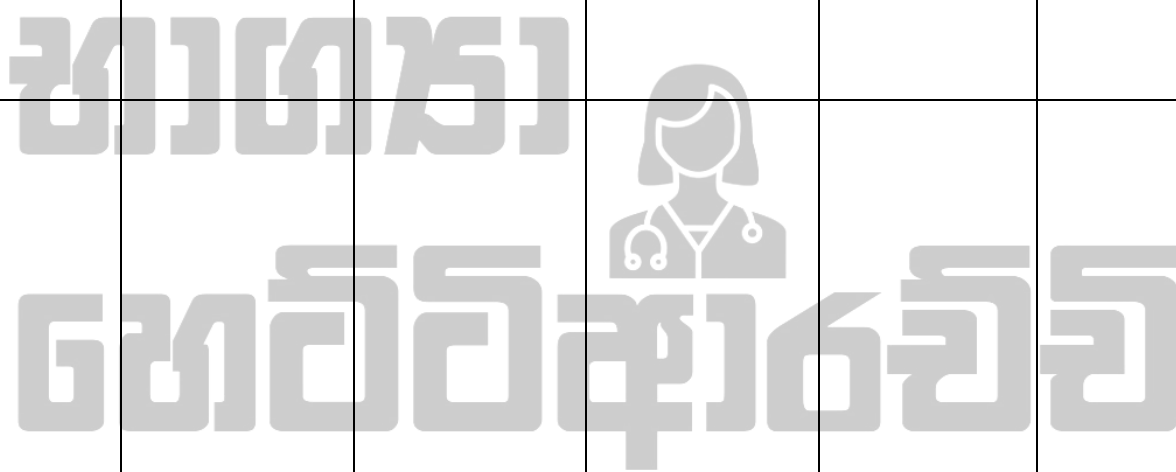
(i) .



(ii).

ආරක්ෂා
සෞඛ්‍ය ආරක්ෂක



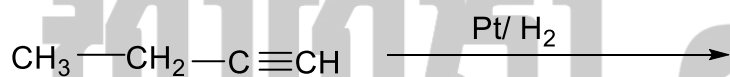
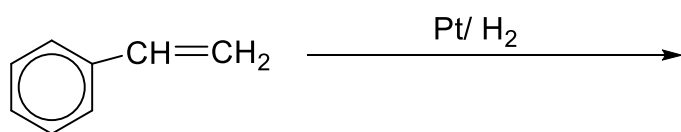
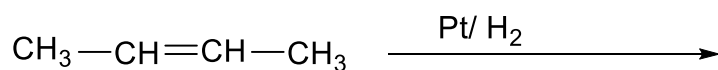
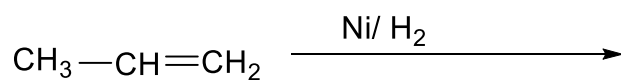
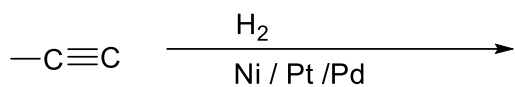
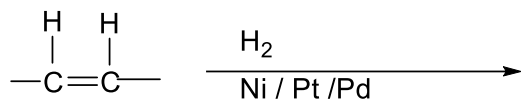


ඇල්කේන	ඇල්කේන	ඇල්කේන	
			බන්ධන ස්වභාවය
			පොදු අණුක සූත්‍රය
			තාපාංක/ද්‍රවාංක චිවලනය
			ලාක්ෂණික ප්‍රතික්‍රියා වර්ගය (යාන්ත්‍රිකය)
			වෙනත්

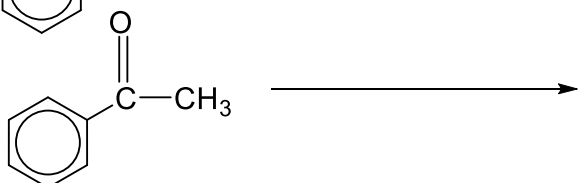
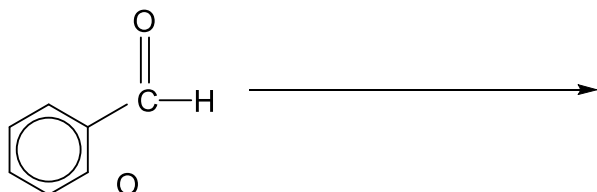
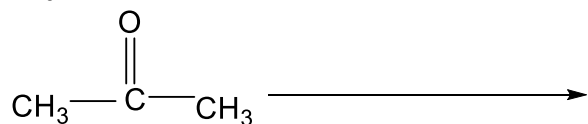
භාෂ්‍යා හේට්ට්ටියාරච්චි

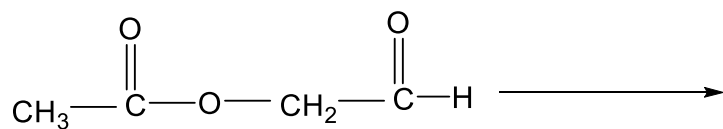


(1) හයිඩ්‍රජනීකරණය



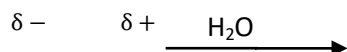
(2) ඇල්ඩිහයිඩ් හා කීටෝන මගින් (ක්ලෙමෙන්සන් ඔක්සිකරණය)



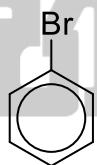
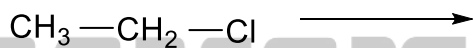
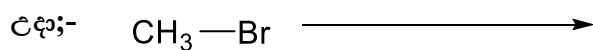
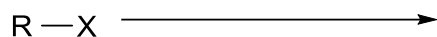


(3) ශ්‍රිතාඨ ප්‍රතිකාරකය මගින්

ශ්‍රිතාඨ ප්‍රතිකාරකය

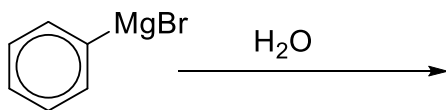


ශ්‍රිතාඨ ප්‍රතිකාරකය නිපදවීම $\longrightarrow$



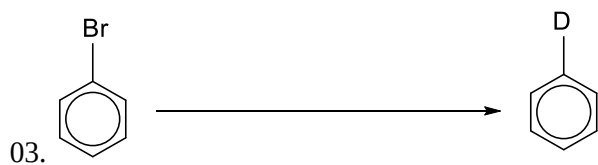
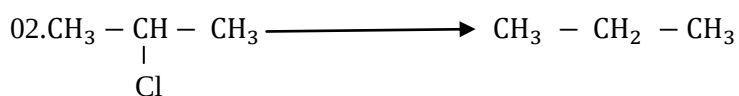
NOTE :

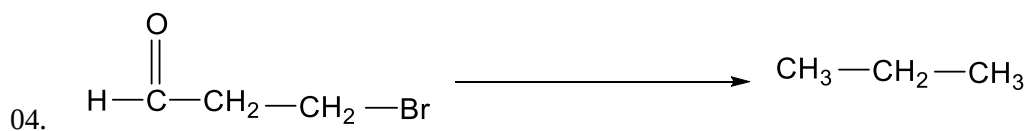




Special Note :

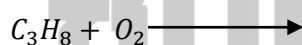
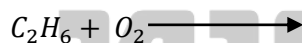
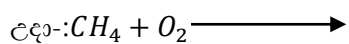
ශ්‍රීතාපිති C ගණනට වඩා වැඩි C ගණනක් ඇති ඇල්කේන නිපදවීම.





ඇල්කේන වල ප්‍රතික්‍රියා

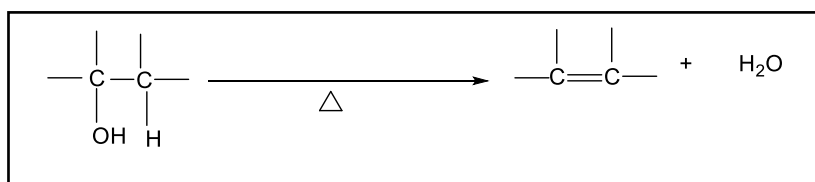
01. O_2 සමඟ



02.

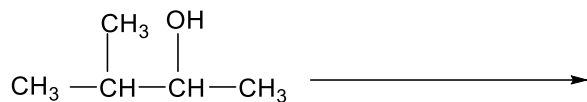
ඇල්කීන නිපදවීමේ ක්‍රම

01. ඇල්කොහොල මගින්

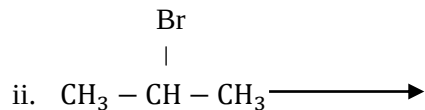
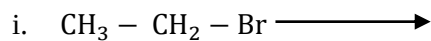


Saytzeffගේ ඉවත් වීමේ නියම -:

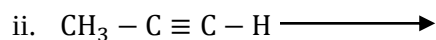
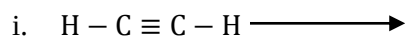
-OH බැඳුණු C ට යාබද C පරමාණු කිහිපයක අසමාන H පරමාණු ගණනක් ඇති විට H අඩුවෙන් පවතින පරමාණුවෙන් H ඉවත් වේ.



02. ඇල්කයිල් හේලයිඩ් මගින්

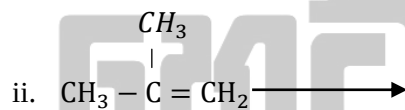
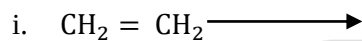


03. ඇල්කයින් මගින්

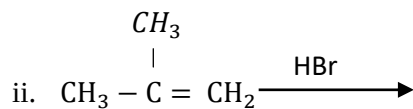
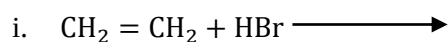


ඇල්කීන් වල ප්‍රතික්‍රියා

01. H_2 සමග (හයිඩ්‍රජනීකරණය)

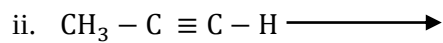


02. HBr සමග

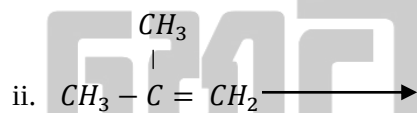
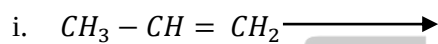
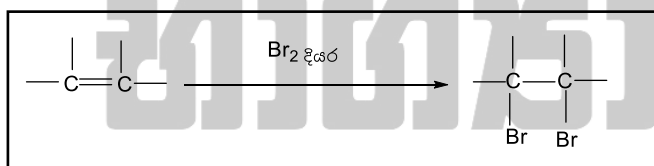


ඇල්කීන් වල ආකලන ප්‍රතික්‍රියා වල සීග්‍රතාවය

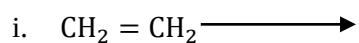
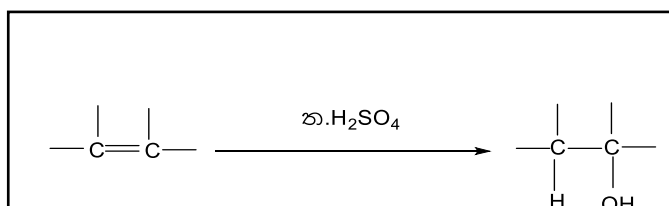
පරිවර්තන

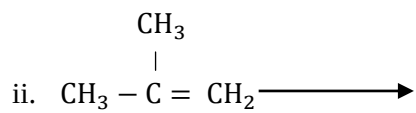


03. Br_2 (දියර සමඟ)

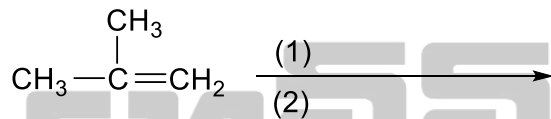
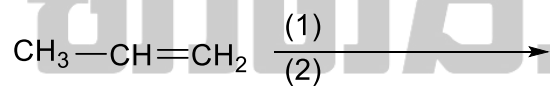


04. තනුක H_2SO_4 සමඟ

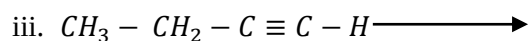
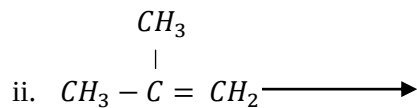
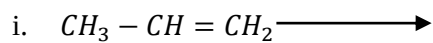
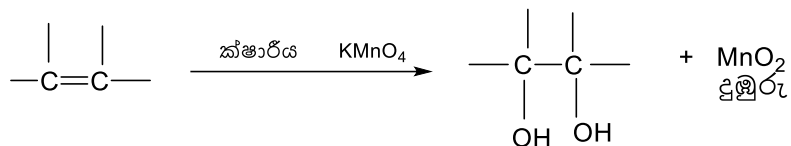




සාන්ද්‍ර H_2SO_4 සමඟ ප්‍රතික්‍රියා කරවා ලැබෙන ඵලය ජලවිච්ඡේදනය කිරීම.



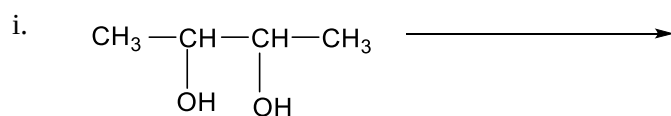
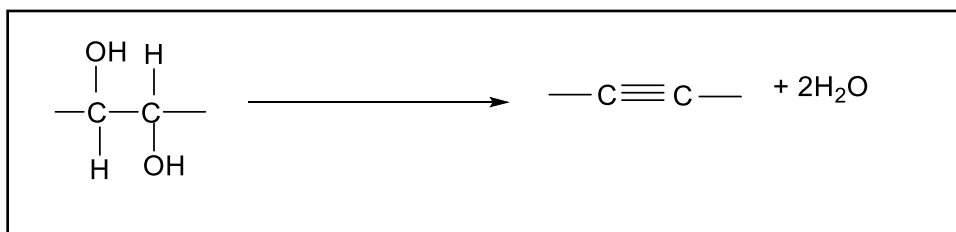
05. ක්ෂාරීය KMnO_4 සමඟ ($\text{KMnO}_4/\text{OH}^-$)



ඇල්කීන හඳුනා ගැනීම

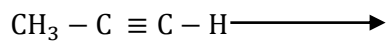
01.
02.
03.

ඇල්කයීන නිපද වීම

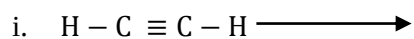
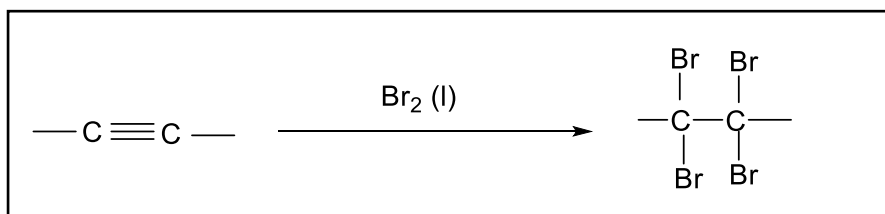


ඇල්කයින් වල ප්‍රතික්‍රියා

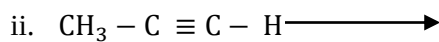
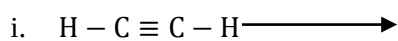
01. H_2 සමඟ



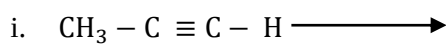
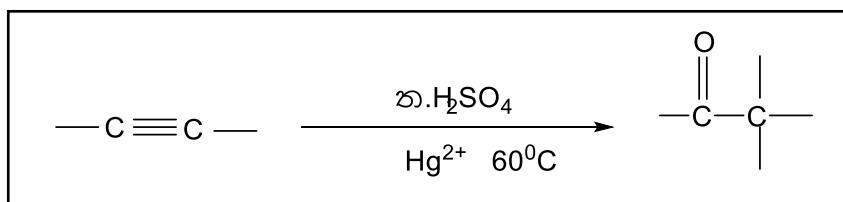
02. Br_2 සමඟ

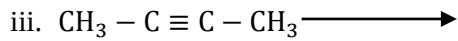
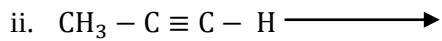


03. HBr සමඟ

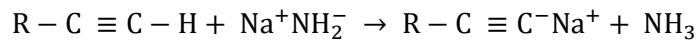


සජලනය

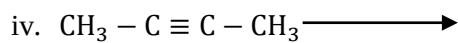
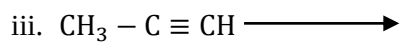
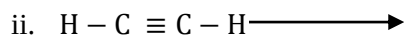




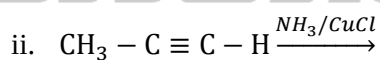
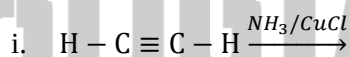
ii. NaNH_2 සමඟ ($\text{Na}^+ \text{NH}_2^-$)



iii. ඇමෝනියම් සිල්වර් නයිට්‍රේට් සමඟ



iv. ඇමෝනියම් කියුප්‍රස් ක්ලෝරයිඩ් (CuCl/NH_3) (Cu_2Cl_2)



ඇල්කයින් හඳුනා ගැනීම

01. අසංතෘප්තතාවය මගින් හඳුනා ගැනීම

02. ආම්ලික H මගින් හඳුනා ගැනීම

I.

II.

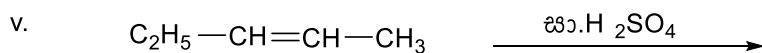
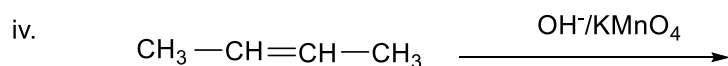
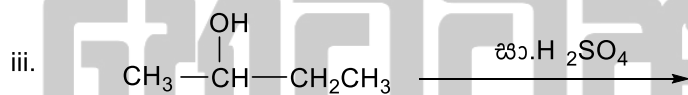
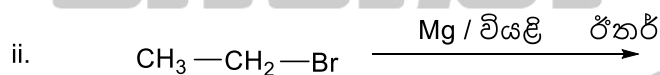
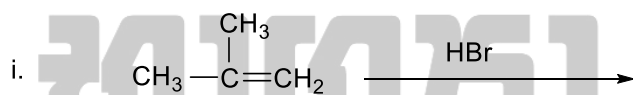
III.

IV.

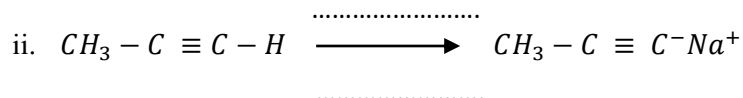
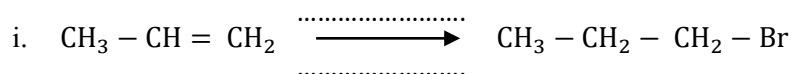
හයිඩ්‍රොකාබන හඳුනා ගැනීමේ සාරාංශ සටහන

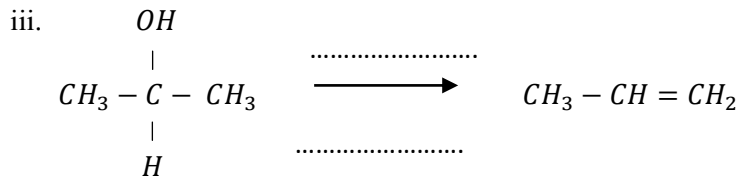
	$CH_3 - CH_3$	$CH_2 = CH_2$	$CH_3 - C \equiv CH$	$CH_3 - C \equiv C - CH_3$
Br_2 දියර යෙදීම				
$H^+ / KMnO_4$				
$OH^- / KMnO_4$				
$NH_3 / AgNO_3$				
$NH_3 / CuCl$				

01. පහත සඳහන් ප්‍රතික්‍රියා වලදී සෑදෙන ප්‍රධාන ඵලවල ව්‍යුහ අඳින්න.



02. පහත සඳහන් ප්‍රතික්‍රියා සඳහා යෝග්‍ය වන ප්‍රතිකාරක හා තත්ත්ව ලියා දක්වන්න.





පහත සඳහන් වගුව සම්පූර්ණ කරන්න.

ප්‍රතික්‍රියක	ප්‍රතිකාරකය සහ තත්ත්ව	සක්‍රීය ප්‍රභේදය	ප්‍රධාන ඵලය	ප්‍රතික්‍රියා වර්ගය
$CH_3 - CH = CH_2$	HBr			
$CH_3 - CH = CH_2$	ක. H_2SO_4			
$CH_3 - CH = CH_2$	Br_2			
$CH_3 - CH - CH_3$ $ $ OH	නිර්ජලීය. $Al_2O_3\Delta$			

ඇරෝමැටික හයිඩ්‍රොකාබන්

සරලතම ඇරෝමැටික හයිඩ්‍රොකාබන් වන්නේ බෙන්සීන් ය. (C_6H_6)

බෙන්සීන් - C_6H_6

- අණුක සූත්‍රය -
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කෙතෙල් ව්‍යුහය

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බෙන්සීන් වල සම්ප්‍රයුක්ත ව්‍යුහ

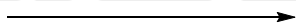
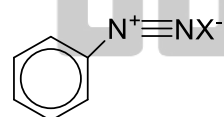
බෙන්සීන් නිපදවීම

1. ෆීනෝල් මගින්

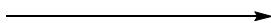
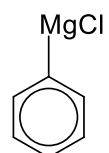
2. OH



2 බෙන්සීන් වල ඩයිසෝනියම් ලවණ මගින්



3. ශ්‍රිනාඩ් මගින්



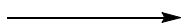
බෙන්සීන් වල ප්‍රතික්‍රියා

ප්‍රධාන වර්ග දෙකකි.

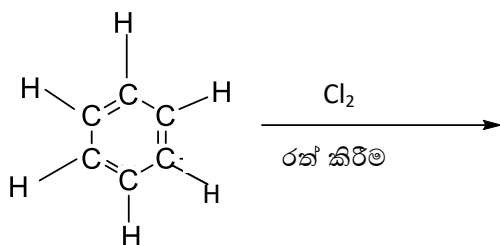
1.
2.

බෙන්සීන්වල ආකලන ප්‍රතික්‍රියා සමග

H₂ සමග

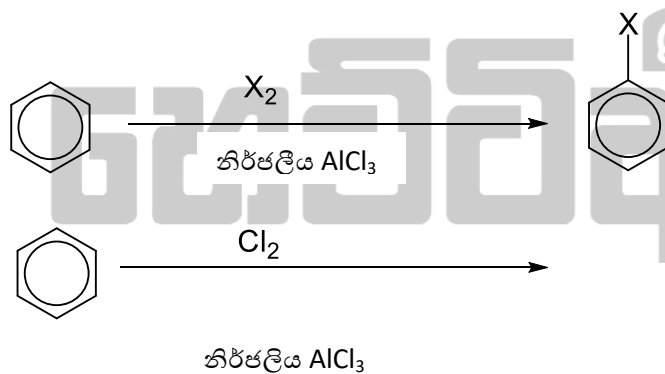


Cl₂ සමග

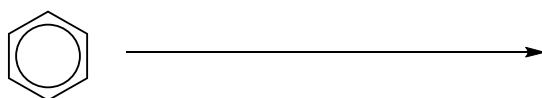


බෙන්සීන් වල ඉලෙක්ට්‍රොෆිලික ආදේශ ප්‍රතික්‍රියා

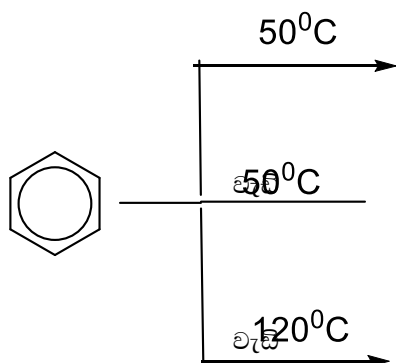
1. හැලජනීකරණය



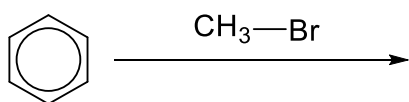
2. නයිට්‍රොකරණය



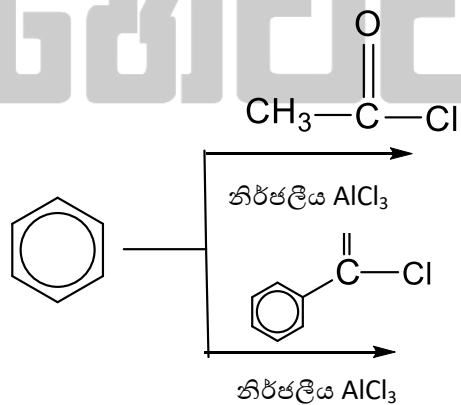
උෂ්ණත්වය මත සෑදෙන ප්‍රතිඵල වෙනස්වේ.



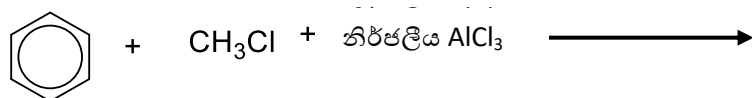
3. ශ්‍රිඩල්ක්‍රාග්ටි ඇල්කිල්කරණය



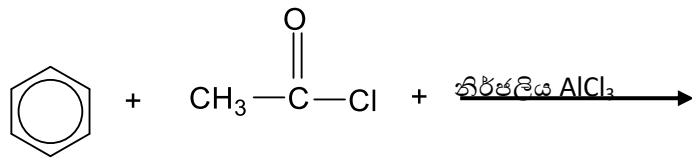
4. ශ්‍රිඩල්ක්‍රාග්ටි ඇසිල්කරණය



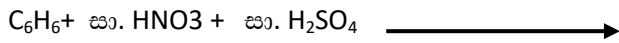
බෙන්සීන් ඇල්කිල්කරණය



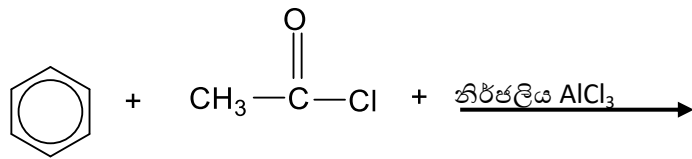
බෙන්සීන් ඇසිල්කරණය



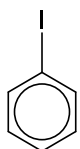
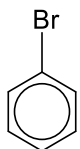
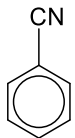
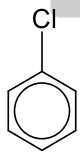
බෙන්සීන් නයිට්‍රොකරණය



බෙන්සීන් ඇසිල්කරණය



බෙන්සීන් සංයෝග නාමකරණය



ඇරෝමැටික සංයෝග වල යොමුකාරක ගුණය

ඇරෝමැටික සංයෝග වල යොමුකාරක වර්ග දෙකකි.

- I.
- II.

❖ Ortho – Para යොමුකාරක

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❖ Meta යොමුකාරක

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❖ බෙන්සීන් වල සක්‍රීය කාණ්ඩ

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❖ බෙන්සීන් වල වික්‍රීය කාණ්ඩ

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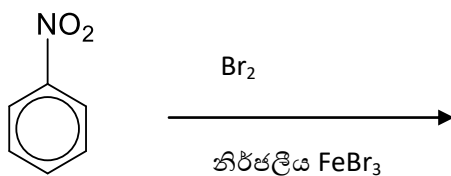
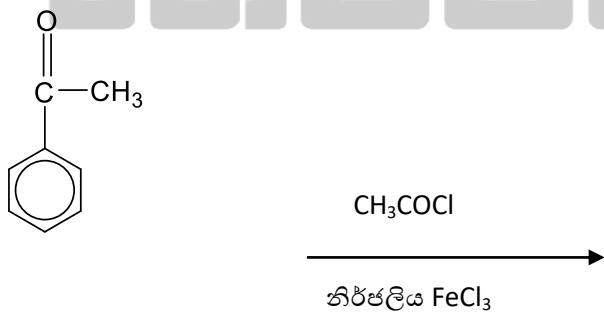
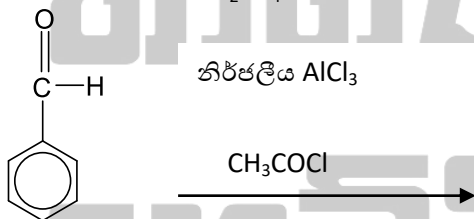
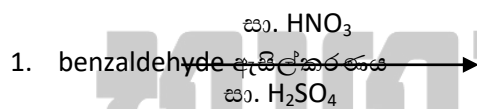
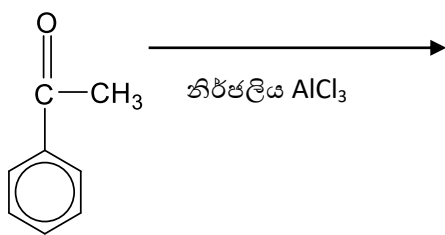
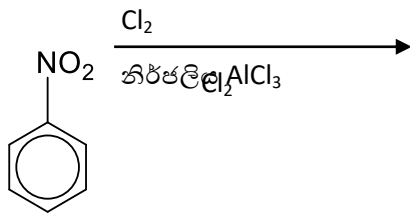
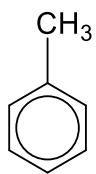
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පහත සඳහන් ප්‍රතික්‍රියාවලදී ලැබෙන එල මොනවාද?

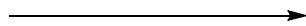
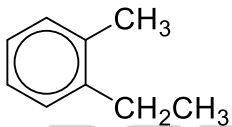
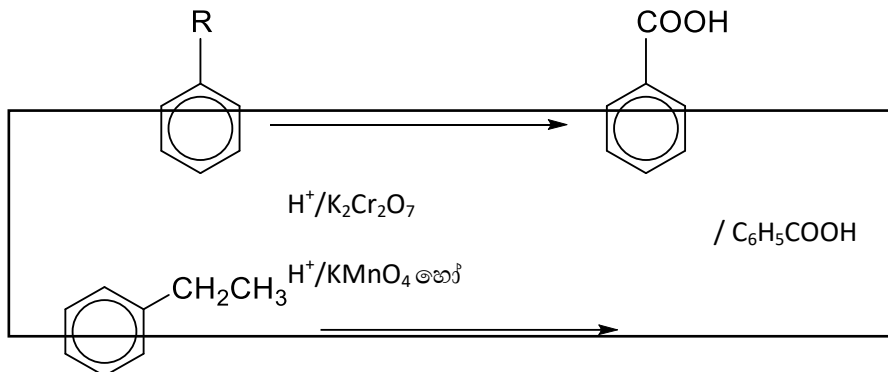


Methylbenzene (toluene)

අණුක සූත්‍රය -



අල්කයිල් බෙන්සින් සෑදීම සඳහා AlBr_3



පහත පරිවර්තනය සිදු කරන්න.

